

OLDENBURG, Germany

WMO: 102150

Lat: 53.183N

Lon: 8.167E

Elev: 66

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	20.0	10.7	9.5	18.8	14.8	11.7	22.0	27.4	43.4	23.8	41.7	7.5	90	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	84.9	66.8	80.8	65.7	76.9	63.7	68.8	80.1	67.0	77.2	65.5	74.5	8.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.0	72.6	63.3	87.4	70.7	61.9	82.9	69.3	32.9	80.2	31.5	77.4	30.3	74.5	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.3	16.1	DB	11.0	91.4	5.1	3.1	7.4	93.6	4.4	95.4	1.6	97.2	-2.1	99.4	(4)
(5)			WB	10.4	72.5	5.0	1.6	6.8	73.7	3.9	74.6	1.1	75.5	-2.6	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	36.0	36.9	41.4	48.7	55.2	60.1	64.9	64.6	58.1	50.4	42.9	36.0	(6)	
	DBStd	12.41	8.13	7.77	6.47	6.94	6.21	5.96	5.90	5.48	5.23	6.35	6.51	8.21	(7)	
	HDD50	1935	434	369	272	105	21	0	0	0	2	73	225	434	(8)	
	HDD65	5769	898	788	731	489	309	171	77	76	216	452	664	898	(9)	
	CDD50	1821	1	2	7	67	183	304	461	451	246	86	11	2	(10)	
	CDD65	177	0	0	0	1	6	25	73	62	10	0	0	0	(11)	
	CDH74	1496	0	0	0	15	73	237	615	482	72	2	0	0	(12)	
	CDH80	419	0	0	0	0	10	61	193	148	7	0	0	0	(13)	
(14) Wind	WSAvg	7.4	8.5	8.8	8.9	7.6	7.2	7.0	6.6	6.4	6.5	7.0	7.1	7.5	(14)	
(15) Precipitation	PrecAvg	32.0	2.8	2.3	2.2	1.7	2.4	2.9	3.4	3.0	2.9	2.5	2.8	3.1	(15)	
	PrecMax	44.3	5.5	4.6	4.8	3.9	4.3	6.1	7.1	6.9	7.6	7.6	5.7	7.0	(16)	
	PrecMin	24.6	0.2	0.6	0.4	0.1	0.8	0.9	1.1	0.9	0.6	0.5	0.1	0.7	(17)	
	PrecStd	4.7	1.5	1.1	1.3	0.9	1.0	1.4	1.5	1.3	1.6	1.3	1.4	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	55.8	63.2	76.3	81.2	86.4	90.6	89.6	80.7	69.9	59.9	55.0	(19)	
		MCWB	50.4	49.6	51.5	57.2	63.9	67.4	68.1	67.7	65.4	61.8	54.3	52.3	(20)	
	2%	DB	51.5	52.5	58.1	70.2	75.5	80.6	85.3	84.4	75.4	65.0	56.6	52.0	(21)	
		MCWB	48.4	48.4	50.0	55.6	60.0	64.9	67.8	66.0	64.4	58.6	53.6	49.7	(22)	
	5%	DB	48.6	49.8	54.4	65.5	71.2	75.7	80.9	79.3	70.6	61.8	54.2	49.4	(23)	
		MCWB	46.1	46.9	48.6	53.4	58.4	63.0	66.4	65.8	61.4	57.0	51.1	46.9	(24)	
	10%	DB	46.4	47.4	51.5	61.0	67.4	71.2	76.5	75.0	66.8	59.3	51.7	47.1	(25)	
		MCWB	44.1	44.6	46.6	51.6	56.3	61.3	64.3	64.1	59.3	55.3	48.9	45.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	51.6	54.2	59.8	65.8	69.0	72.0	71.1	67.4	61.9	56.0	52.5	(27)	
		MCDB	53.3	54.5	60.1	69.8	77.3	81.9	84.1	83.0	77.1	68.4	58.0	54.3	(28)	
	2%	WB	48.8	49.0	51.5	56.8	62.3	66.7	69.4	68.2	64.5	59.6	53.7	49.9	(29)	
		MCDB	51.1	51.4	55.6	66.7	72.5	77.3	81.0	78.6	73.8	63.8	56.4	52.0	(30)	
	5%	WB	46.3	46.9	49.4	54.9	59.6	64.7	67.3	66.4	62.4	57.6	51.3	47.3	(31)	
		MCDB	48.2	49.5	53.7	63.4	67.7	72.9	77.7	75.6	69.2	60.8	53.5	49.0	(32)	
	10%	WB	44.2	44.9	47.0	52.6	57.3	62.4	65.5	65.0	60.4	55.9	49.2	45.0	(33)	
		MCDB	46.3	47.4	50.9	60.3	65.4	69.6	74.7	73.3	65.2	58.9	51.3	46.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.3	12.6	16.6	17.0	16.2	16.1	16.1	14.5	12.6	8.8	7.7	(35)	
		MCDBR	8.9	10.3	17.7	24.3	24.1	23.2	24.1	23.3	19.9	14.5	9.8	8.8	(36)	
	5% WB	MCWBR	7.8	8.2	11.3	12.2	12.0	10.9	10.0	9.7	10.2	9.4	7.5	8.1	(37)	
		MCWBR	8.8	9.2	15.6	20.6	20.6	20.8	22.0	19.4	18.0	13.3	9.4	8.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.345	0.376	0.409	0.406	0.404	0.419	0.412	0.387	0.360	0.339	0.318	(40)	
		taud	2.323	2.326	2.259	2.219	2.262	2.290	2.262	2.296	2.356	2.398	2.352	2.314	(41)	
	Ebn at Noon		203	238	254	260	267	269	262	257	248	229	196	182	(42)	
		Edh at Noon	19	26	34	40	40	39	40	36	30	24	19	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		189	402	771	1275	1580	1663	1587	1321	953	536	233	145	(44)	
	RadStd		20	58	84	144	159	128	203	101	109	55	24	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-230	N/A	N/A	N/A

Nomenclature: See separate page